

Brainstorming & Ideation

Project: FitTrack AI - Intelligent Fitness Tracking & Health Insights Platform

1. PROBLEM IDENTIFIED

- Users need an effective way to track their daily fitness activities, workouts, and health metrics.
- Raw workout logs and physical metrics (e.g., duration, activity types, caloric expenditure) do not always give users actionable insights or clear interpretations of their fitness progress.
- Users benefit significantly from concise health summaries and practical recommendations regarding workout routines, daily targets, and fitness guidance.
- A secure mechanism is required to store user fitness logs, maintain community hub interactions, and protect private user operations.

2. PROPOSED IDEA

FitTrack AI is a modular MVC-style RESTful backend API designed to manage fitness activity tracking, user authentication, personal health logs, and community hub interactions. It integrates AI engines to deliver personalized health summaries and actionable workout guidance while utilizing local rule-based fallback logic to guarantee high service reliability during external network interruptions.

3. TARGET USERS

- **Registered Users:** Fitness enthusiasts who want to record activity logs, track personal health metrics, participate in community discussions, and receive personalized AI health insights.
- **Public API Consumers:** Unauthenticated users or integrations requiring basic system status checks and public endpoint access.

4. CORE IDEA AND SCOPE

Area	Planned Capability Supported by the Project
Authentication	Registration, login, profile access, bcrypt password hashing, and JWT authorization.
Activity Management	Create, view, update, and delete (CRUD) fitness logs, workout entries, and user activities.
Community Hub	Post sharing, interaction management, and community activity tracking.
AI Health Insights	

Area	Planned Capability Supported by the Project
	Generate concise health summaries, progress reports, and practical workout recommendations.
Reliability	Use local rule-based fallback mechanisms when external AI services or network calls are unavailable.

5. EXPECTED OUTCOME

The result is a structured, scalable RESTful backend API built with Node.js, Express.js, and MongoDB that manages user authentication, fitness activity logging, and community features while providing AI-generated fitness insights.

Scope note: The system strictly focuses on fitness activity tracking, user account management, health recommendations, and community hub features as implemented in the FitTrack AI project. Unrelated functional modules outside this scope are excluded.